

Virtual Stereo Vision-Based Localization and 3D Mesh Mapping for Dual UAV Systems

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Abstract—Simultaneous Localization and Mapping (SLAM) has been extensively applied to solve the problem of robot localization and mapping without absolute position measurements. Visual sensors are widely used in odometry techniques due to their low cost and richer information content. However, when robots operate far from the scene (e.g., high-altitude UAVs), visual-inertial odometry (VIO) positioning accuracy degrades significantly. This compromises the balance between autonomous precision and rapid scene understanding. To address this issue, we propose a collaborative dual-UAV localization and real-time scene reconstruction framework for high-altitude operations. By utilizing inter-UAV visual overlap observations and wireless ranging constraints, we propose a collaborative decentralized trajectory localization and global map point consistency optimization scheme. Our solution achieves high-altitude precise positioning and maintains dual-UAV observation consistency, enabling real-time construction of large-scale navigable maps. Evaluation results in photorealistic simulated environments demonstrate the superior accuracy of our approach in high-altitude localization and scene reconstruction.

I. INTRODUCTION

Autonomous navigation is essential for unmanned systems to perform complex tasks. It requires two key technologies: precise localization and environmental mapping. When operating in unknown environments, unmanned aerial vehicles (UAVs) must instantaneously and accurately estimate their positions while perceiving surrounding structures to enable real-time path planning and collision avoidance. Although GNSS (e.g., GPS/RTK) provides reliable positioning in open areas [1], signal degradation and localization drift occur in high-altitude, indoor, or complex environments [2]. Moreover, GNSS systems lack the capability to provide 3D environmental geometry, further complicating autonomous navigation in cluttered or GPS-denied spaces.

To overcome these limitations, visual SLAM (Simultaneous Localization and Mapping) has emerged as a viable solution. By leveraging onboard cameras and sensors, visual SLAM enables GPS-denied autonomous localization and environmental mapping [3]. In recent years, visual-inertial odometry (VIO) SLAM technique that fuses visual and inertial data has become the de facto standard for UAV navigation due to its cost-efficiency, low power consumption,

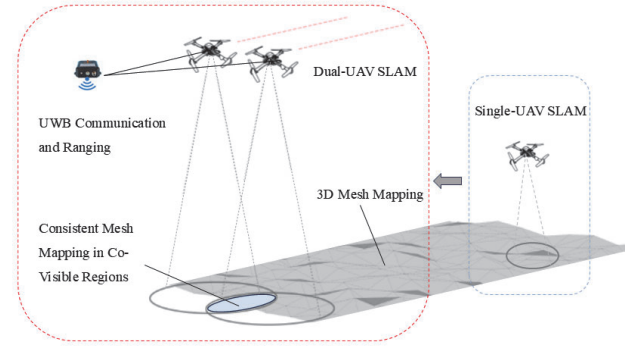


Fig. 1. Dual UAVs perform autonomous localization and real-time scene mapping at higher altitudes relative to the scene in large-scale environments. Wireless communication is utilized for distance measurement and information transmission between the two UAVs, achieving consistent map construction through co-observation of ground scenes.

and high precision [4]. However, while VIO exhibits exceptional accuracy in close-range scenarios (e.g., low-altitude flights), its performance deteriorates significantly in large-scale environmental mapping.

In high-altitude flight scenarios, although the camera's field of view (FOV) expands, the increased scene distance introduces severe scale drift and feature sparsity, substantially degrading VIO's localization and mapping accuracy [5]. During high-altitude flight, the greater scene depth leads to narrower triangulation angles, significantly increasing depth estimation uncertainty [6]. In monocular VIO systems, the global scale depends entirely on IMU measurement integration. To achieve adequate triangulation angles, the UAV must execute larger baseline movements, which amplifies scale drift due to error accumulation in inertial data.

To improve positioning accuracy and mapping quality in high-altitude flight scenarios, we propose an optimized method using a dual-UAV system. The cooperative effect between two peer UAVs is utilized to increase the baseline length between them, thereby optimizing triangulation angles and significantly improving positioning accuracy. Simultaneously, by fusing point cloud data collected from the expanded observation ranges of both UAVs, more complete and detailed scene mesh maps can be rapidly generated, thereby providing more reliable references for unmanned system navigation and obstacle avoidance as shown in Fig. 1. This method offers stronger support for autonomous navigation of UAVs in complex environments.

The main contributions of this paper can be summarized

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as follows:

- 1) We present a real-time lightweight visual SLAM method designed for dual-UAV systems. This approach enhances positioning accuracy in high-altitude scenarios through effective fusion of data from both UAVs, while enabling fast and precise construction of consistent mesh maps in large-scale environments.
- 2) We develop an efficient dual-UAV map consistency fusion algorithm. This algorithm intelligently integrates data from both UAVs to produce high-quality, complete scene mesh maps, thereby substantially improving mapping accuracy and operational efficiency.
- 3) Through comprehensive experiments, we validate the effectiveness and superiority of our proposed method. The approach demonstrates exceptional positioning accuracy and rapid mesh map construction capability in high-altitude, large-scale environments, confirming its practical potential for navigation and obstacle avoidance tasks.

II. RELATED WORK

Multi-robot SLAM (Simultaneous Localization and Mapping) has emerged as a prominent research focus, with its core objective being collaborative environment mapping and self-localization among multiple robots[7]. To address the degradation of VIO performance at high altitudes, prior work [8] demonstrated significant VIO results at elevations up to 50 meters. Alternative approaches employ dual-camera setups mounted at opposite ends of fixed-wing aircraft to extend the baseline of virtual stereo cameras [9][10], though such methods remain constrained by the physical dimensions of the airframe. Studies in [11][12] utilize two independent UAVs to construct shared maps with information consistency constraints, effectively enhancing both localization accuracy and mapping efficiency in multi-robot systems. However, in complex environments, inter-robot communication often proves unstable, prompting investigations in [13][14] into distributed filtering algorithms that enable efficient data exchange and cooperative localization under limited communication bandwidth.

Point cloud-based scene reconstruction has gained widespread adoption in robotic navigation due to its ability to provide precise geometric representations of environments. Recent advances in Neural Radiance Fields (NeRF) and 3D Gaussian Splatting [15][16] have significantly improved reconstruction quality, though their high computational costs pose challenges for real-time implementation. [17] pioneered real-time semantic mesh reconstruction, while [18] extended this capability to multi-robot systems, albeit with notable scale drift issues in large-scale environments. The [19] framework introduced an innovative incremental mesh reconstruction approach using LiDAR sensors, achieving substantial improvements in both reconstruction speed and accuracy across extensive areas.

Inspired by [12], we employ a dual-drone system relying solely on camera sensors (without LiDAR) to achieve precise

localization in high-altitude areas. By leveraging visual co-observation overlap constraints for map consistency, our approach enables faster and more accurate large-scale mapping compared to single-drone low-altitude mapping.

III. METHOD

A. System Overview

The distributed SLAM system is implemented on two identical UAVs that communicate via radio links. As shown in Fig.3 (execution module of one UAV), each drone's functionality is divided into two core modules: (1) Localization Module - primarily consisting of MSCKF-based sensor fusion positioning and inter-UAV ADMM optimization; (2) Mapping Module.

Localization Module: The localization module is responsible for estimating the initial pose of each image frame by fusing data from inertial sensors and camera sensors using an Extended Kalman Filter (EKF). It selects representative images over a period of time as keyframes (KFs); Upon obtaining the poses of these keyframes and their observed map points, an optimization thread is initiated. This thread performs map point matching in the co-visible areas between the two agents. The optimization incorporates the current keyframe information from the local agent, the latest received keyframe information from the other UAV, UWB ranging data, and co-visibility information to refine both frame poses and map point locations. The optimized pose deviations serve as pseudo-observations for the EKF to correct trajectory positions.

Mapping Module: In the mapping module, the consistent camera poses and map point positions stored by both UAVs are used as input. To enable incremental mapping in large-scale environments, four data types are employed to represent different scene structures: mesh vertices, triangular faces, regions, and voxels. Two data structures(a hash table and an octree)are used for data storage and processing. The global mesh map is constructed through dimensionality reduction projection of vertices, Delaunay triangulation and spatial inverse transformation.

B. MSCKF-based pose estimation

To utilize the optimized deviations for pose refinement of non-keyframes, we designed an MSCKF to fuse inertial sensor data, camera sensor data, and pseudo-observation information based on optimization deviations. While incorporating map points (MPs) can improve the robustness and accuracy of the filter, it also leads to excessive computational complexity. Therefore, we only add a small number of repeatedly observed feature points to the filter.

The state representation is similar to other MSCKF [20] as follows:

$$\mathbf{x}_k = [\mathbf{x}_I^\top \quad \mathbf{x}_C^\top \quad \mathbf{x}_M^\top \quad \mathbf{x}_W^\top \quad c_{t_I}]^\top \quad (1)$$

$$\mathbf{x}_I = \begin{bmatrix} I_k \bar{q}^\top & G \mathbf{p}_{I_k}^\top & G \mathbf{v}_{I_k}^\top & \mathbf{b}_{\omega_k}^\top & \mathbf{b}_{a_k}^\top \end{bmatrix} \quad (2)$$

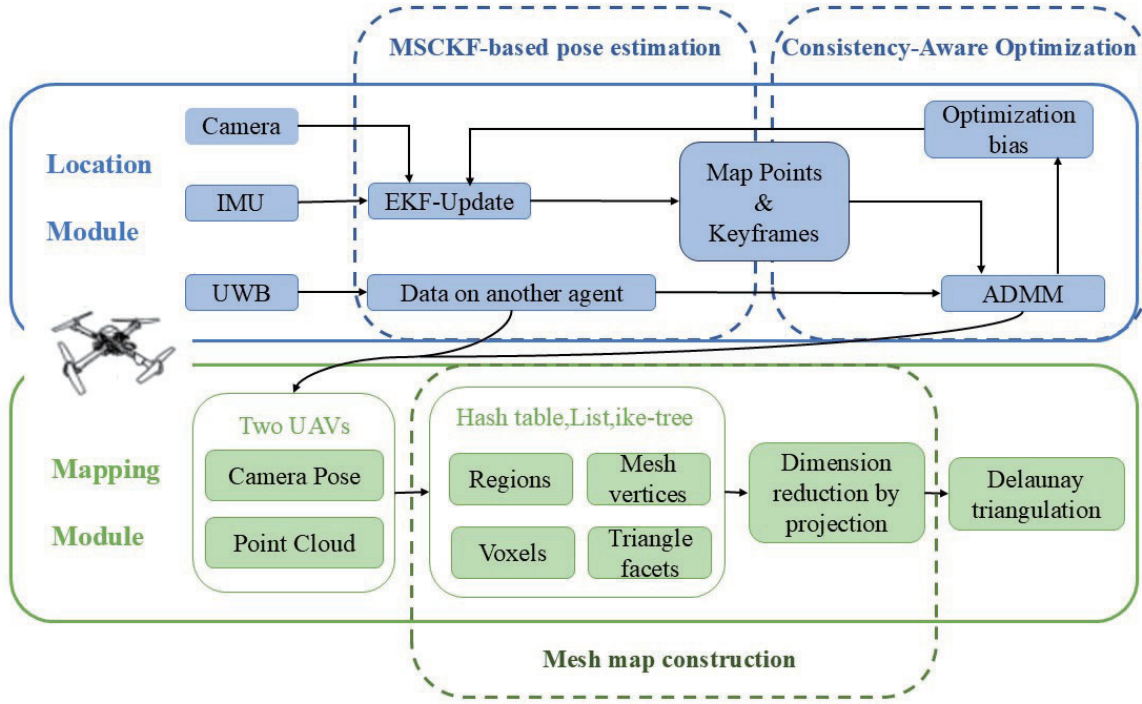


Fig. 2. The overview of our proposed system. The system runs on two identical agents that communicate via wireless transmission for data exchange and ranging. Each agent consists of two main modules: a localization module and a mapping module.

$$\mathbf{x}_C = \begin{bmatrix} I_{k-1}^c \bar{q}^\top & G \mathbf{p}_{I_{k-1}}^\top & \cdots & I_G^{k-c} \bar{q}^\top \end{bmatrix}^\top \quad (3)$$

$$\mathbf{x}_M = \begin{bmatrix} G \mathbf{p}_{f_1}^\top & \cdots & G \mathbf{p}_{f_m}^\top \end{bmatrix}^\top \quad (4)$$

$$\mathbf{x}_W = \begin{bmatrix} I_G^c \bar{q}^\top & c_1 \mathbf{p}_I^\top & \zeta_0^\top & \cdots & \zeta_w^\top \end{bmatrix}^\top \quad (5)$$

Where $\mathbf{x}_I$ represents IMU-related states, $\mathbf{b}_{\omega_k}/\mathbf{b}_{a_k}$ denote accelerometer and gyroscope biases respectively, $G \mathbf{p}_{I_k}^\top$ and $G \mathbf{V}_{I_k}^\top$ indicate the position and velocity of the IMU in the global reference frame. $\mathbf{x}_C$ represents the pose of the c -th camera. $\mathbf{x}_M$ denotes the m -th environmental landmark. $\mathbf{x}_W$ includes intrinsic and extrinsic camera parameters: focal length, optical center, distortion parameters, and camera-IMU extrinsics. To account for synchronized camera clocks, the calibration vector incorporates a single time offset between IMU and camera clocks.

The IMU bias variables are modeled as white Gaussian noise processes over time. State estimation is performed based on the current estimates as follows:

$$\hat{\mathbf{x}}_{k|k-1} = f(\hat{\mathbf{x}}_{k-1|k-1}, {}^I \mathbf{a}_m, {}^I \omega_m, \mathbf{0}) \quad (6)$$

The IMU bias variables are modeled as white Gaussian noise processes, where ${}^I \mathbf{a}_m$ and ${}^I \omega_m$ represent the IMU's linear acceleration and angular velocity respectively, $\hat{\bullet}$ denotes the estimated value, the subscript $k | k-1$ indicates the predicted estimate at time k given measurements up to

time $k-1$, and the state covariance matrix $\mathbf{P}$ is propagated by linearizing the nonlinear model at the current estimate:

$$\mathbf{P}_{k|k-1} = {}_{k-1} \mathbf{P}_{k-1|k-1} {}_{k-1}^\top + \mathbf{Q}_{k-1} \quad (7)$$

where ${}_{k-1}$ is the state transition Jacobian matrix and $\mathbf{Q}_{k-1}$ is the process noise covariance matrix [21].

The optimized errors need to be synchronized to the estimates of non-keyframes, thus they are modeled as pseudo-observations and incorporated into the EKF. This is achieved by merging the real observations and pseudo-observations into a new augmented measurement matrix.

$$\tilde{\mathbf{z}}_k = \begin{bmatrix} \mathbf{z}_k^\top, \mathbf{z}_k^{\text{PT}^\top} \end{bmatrix}^\top \quad (8)$$

$$\tilde{\mathbf{H}}_k = [\mathbf{H}_k^\top, \mathbf{H}_k^{\text{PT}^\top}]^\top \quad (9)$$

$$\tilde{\mathbf{R}}_k = \begin{bmatrix} \mathbf{R}_k & \mathbf{0} \\ \mathbf{0} & \mathbf{R}_k^p \end{bmatrix} \quad (10)$$

where $\tilde{\bullet}$ represents the update incorporating pseudo-observations. A standard MSCKF update is performed as follows:

$$\hat{\mathbf{x}}_{k|k} = \hat{\mathbf{x}}_{k|k-1} \boxplus \mathbf{K}_k (\tilde{\mathbf{z}}_k - h(\hat{\mathbf{x}}_{k|k-1})_{k|k-1}) \quad (11)$$

$$\mathbf{P}_{k|k} = \mathbf{P}_{k|k-1} - \mathbf{K}_k \tilde{\mathbf{H}}_k \mathbf{P}_{k|k-1} \quad (12)$$

$$\mathbf{K}_k = \mathbf{P}_{k|k-1} \tilde{\mathbf{H}}_k^\top (\tilde{\mathbf{H}}_k \mathbf{P}_{k|k-1} \tilde{\mathbf{H}}_k^\top + \tilde{\mathbf{R}}_k)^{-1} \quad (13)$$

where $\boxplus$ represents the nonlinear update operation on the manifold state [22].

C. Consistency-Aware Optimization

The optimization thread plays a crucial role in trajectory refinement for the agents and maintaining global point cloud consistency. It optimizes their trajectories using co-visibility observations between the two agents and UWB ranging measurements as constraints [23]. In cases where the two UAVs lack shared visual observations and communication links owing to various conditions, they will proceed with independent localization and flight as separate, individual UAV systems. To enable each agent to independently construct a complete yet lightweight global mesh map, each agent maintains its own separate map points (MPs) set. To ensure consistency between the agents' observations, the following constraints are introduced:

$$p_{m,i} = p_{m,j}, \quad \forall m \in (m_1 \dots m_l), \quad (14)$$

$$T_{n,i} = T_{n,j}, \quad \forall n \in (n_1 \dots n_k), \quad (15)$$

Where $p_{m,i}$ represents the position of the co-visible point m observed by both agent i and agent j , and $T_{n,i}$ denotes the pose of keyframe n stored on agent i . As proposed in [24], for each pairwise consistency constraint, a dual variable λ_p and λ_T is introduced for each involved observation point and keyframe respectively.

We can formulate the optimization objective as:

$$\begin{aligned} f(\mathcal{X}_i) := & \sum_{K \in \mathcal{K}} \sum_{M \in \mathcal{M}(k+s)} \delta_c (\mathbf{y}_{K,M}^T \mathbf{W}_r^{k,j} \mathbf{y}_{K,M}^T) \\ & + f_c(u) + f_c(T_i, p_i) \end{aligned} \quad (16)$$

To enforce consistent co-observations between the two agents, matched observation points are reprojected onto their respective keyframes in both agents. Here, $\mathcal{K}$ denote the keyframes generated by the two agents, while $M \in \mathcal{M}(k+s)$ represents all observed and matched points in keyframe $\mathcal{K}$. $f_c(u)$ represents the error between the UWB measured distance and the estimated distance between the two agents. The consensus term $f_c(T_i, p_i)$ enforces equality constraints from (14) and (15), with pose consensus formulated as:

$$\mathbf{f}_c(T_i, p_i) = \sum (\delta T_i)^T \bar{\lambda}_T + \frac{\gamma}{2} \|\delta T_i\|^2 \quad (17)$$

The processing of δT_i is performed on the $\text{SO}(3)$ manifold (refer to [12]).

D. Mesh map construction

1) Data Storage

After jointly optimizing the consistency of poses and map point locations, both drones store all information from each other. Each drone can incrementally construct a scene mesh map for navigation and obstacle avoidance based on all stored observations.

All data is stored in four types: vertices, voxels, triangular patches, and regions. The point clouds from all global observations are stored as mesh vertices $\mathbf{V}$ within each voxel $\mathbf{O}$, where each vertex contains a unique index and its 3D position in the global coordinate system ($\mathbf{v}_i \in \mathbb{R}^3$). Each voxel $\mathbf{O}_i$ has a flag f_o indicating whether new vertices have been added. Each triangular mesh consists of three vertices and a triangular face $\mathbf{T}$, which is defined by the ordered vertex indices, face center point, and normal vector. These triangular faces are stored in regions $\mathbf{R}$, which similarly contain a synchronization flag f_R to indicate their synchronization status.

Meanwhile, as the map scale increases, the number of vertices becomes extremely large, necessitating the maintenance of an incremental kd-tree [25] to enable fast searching of mesh vertices. The primary purposes are: 1) To ensure the distance between any two mesh vertices is greater than a minimum value ξ , thereby maintaining appropriate resolution of the triangular mesh; 2) To allow the vertex expansion operation in our voxel meshing process to erode gaps between adjacent voxels.

For each vertex $\mathbf{V}_{ij} \in \mathbf{O}_i$, search for vertices within a radius d_r of it.

$$\forall \mathbf{V} \in \tilde{\mathbf{O}}_i, \quad \|\text{Pos}(\mathbf{V}) - \text{Pos}(\mathbf{V}_{ij})\| \leq d_r. \quad (18)$$

Vertices found outside $\mathbf{O}_i$ are merged into the voxel's vertex set. Three grid subdivisions are then performed to enable cross-voxel mesh construction, compensating for missing inter-voxel connectivity.

1) Spatial Mesh Partitioning

Performing mesh partitioning directly among 3D vertices in space is highly challenging. Therefore, it is necessary to project the 3D vertices onto a 2D plane for mesh partitioning. However, if the selected 2D plane is suboptimal, the reprojection of the 2D mesh back into 3D space will lose the characteristics preserved in the 2D space, resulting in lower-quality reconstructed maps. For vertices within a smaller region, they tend to fit onto a plane, and the mesh partitioned on this plane, when reprojected into 3D space, retains its mesh properties as much as possible. Similar to other approaches, we apply Delaunay triangulation to vertices projected onto a 2D planes well-established method extensively documented in [17].

IV. EXPERIMENTAL EVALUATION

A. Realistic Synthetic Dataset

For real-world outdoor scenarios, maintaining stable flight with two closely positioned drones is challenging due to the

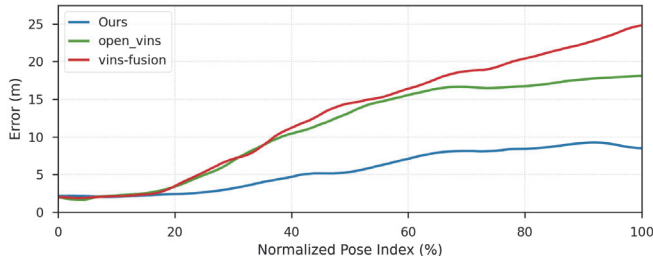


Fig. 3. The localization performance of three algorithms under increasing altitude and distance conditions is compared, clearly demonstrating that the scale-constrained reinforcement method significantly enhances positioning accuracy.

operator's limited field of view, which increases the risk of loss of control. To address this, we employ AirSim [26] to simulate realistic flight sensor data.

To ensure consistency in camera depth and flight altitude, we conduct flights in an almost entirely flat environment. Images are rendered at 640 480 resolution (20 Hz), while IMU data is collected at 400 Hz. UWB relative ranging is simulated by adding Gaussian noise to the true inter-agent distance, with a sampling rate of 20 Hz.

B. Comparative Analysis of Visual-Inertial SLAM Systems at High Altitudes

In this section, we conduct a comparative evaluation with two widely adopted state-of-the-art algorithms, OpenVINS and VINS-Fusion, to assess their autonomous localization accuracy in large-depth scenarios. The experimental data is collected in a simulated environment, where the UAV's trajectory consists of two distinct phases: in the first phase, the UAV ascends vertically from the origin to an altitude of 30 meters, with this endpoint serving as the starting position for the subsequent phase; the second phase employs a helical trajectory with a radius of 100 meters and a vertical ascent of 100 meters per revolution, utilizing a half-circle trajectory (reaching a maximum altitude of 80 meters) as the motion path. During the experiment, two UAVs maintain a formation flight with a constant 10-meter separation. To mitigate the influence of random errors, each algorithm undergoes three independent experimental trials, with the average error calculated accordingly. The resulting mean error after trajectory alignment is presented in Fig.3.

The experimental results indicate that during the initial phase when the UAV's distance to the scene was less than 10m, all algorithms produced estimates closely matching the ground truth values, demonstrating that our method does not necessarily exhibit superior performance over others in shallow-depth scenarios. However, as the UAV altitude increased, both OpenVINS and VINS-Fusion showed progressively larger deviations from the true trajectory, with their errors growing rapidly. Particularly, the VINS-Fusion method ultimately accumulated trajectory errors reaching 8% of the total path length, where the primary source of this error stems from metric scale degradation. In comparison, our approach utilizing an extended virtual stereo

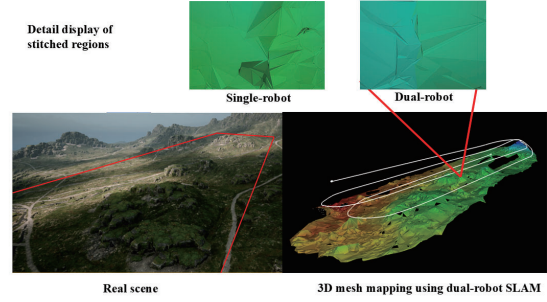


Fig. 4. Large-scale mesh reconstruction is demonstrated using our dual-drone system. Detailed comparisons highlight how co-visibility and ranging constraints enhance mapping consistency, significantly improving accuracy and completeness in stitching regions compared to unconstrained approaches.

camera baseline demonstrated superior performance when facing continuously increasing scene depth. The trajectory error growth remained most gradual throughout the flight, consistently maintaining the lowest error level (over 40% lower than comparative methods at all times). This performance advantage primarily originates from our method's enhanced capability to preserve scale observability under challenging depth conditions through the virtual baseline extension technique, which effectively mitigates the scale drift problem encountered by conventional monocular or short-baseline stereo approaches in large-scale environments. The experimental data clearly shows that while all methods perform comparably in near-range scenarios, our approach maintains significantly better accuracy and stability as the operational range extends, making it particularly suitable for high-altitude UAV applications where maintaining precise metric scale estimation over long distances is crucial for reliable autonomous navigation.

C. Large-Scale Rapid Mapping Performance

We performed grid-based mapping trials using dual UAVs across a 400 m $\times$ 500 m operational area, maintaining a mean altitude of 25 m at a survey speed of 1.5 m/s, achieving full coverage in 30 minutes. Given the computational constraints of real-time camera-based mesh reconstruction for complex geometries, we selected a high-relief yet geometrically sparse test site. Fig.4 presents the reconstructed 3D mesh with quantitative quality metrics.

The experimental results show that in large-scale scenarios, the mapping by dual UAVs can basically reflect the morphology and changes of objects in the scene, and can already provide a simple map foundation for UAV system navigation and obstacle avoidance. Compared with single-UAV mapping at the same appropriate altitude, the dual-UAV system achieves smoother map stitching in co-visible areas and more complete map assembly along reciprocal flight paths. This improvement mainly benefits from the co-visibility constraints of map points between the two UAVs and their more accurate high-altitude positioning. Moreover,

due to the larger observable area covered by dual UAVs, the mapping speed is also superior to that of a single UAV. However, limited by the balance between real-time reconstruction and scene accuracy, there still exist some missing map segments in the scene. This occurs because, to maintain performance balance, we did not perform large-scale feature point extraction and localization. The sparse distribution of feature points leads to reconstructed scenes that can only represent surfaces with significant physical structure changes and relatively simple geometries, while areas with insufficient feature points exhibit mesh gaps.

V. CONCLUSIONS

In this work, we propose a novel framework for autonomous UAV localization and grid-based mapping in high-altitude scenarios. Our approach leverages longer baseline stereo constraints and wireless communication-based ranging between two UAVs to optimize their relative poses and construct a globally consistent map. The system adopts a decentralized optimization strategy, where each UAV independently estimates its trajectory and builds a local map while exchanging key information to maintain global consistency.

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